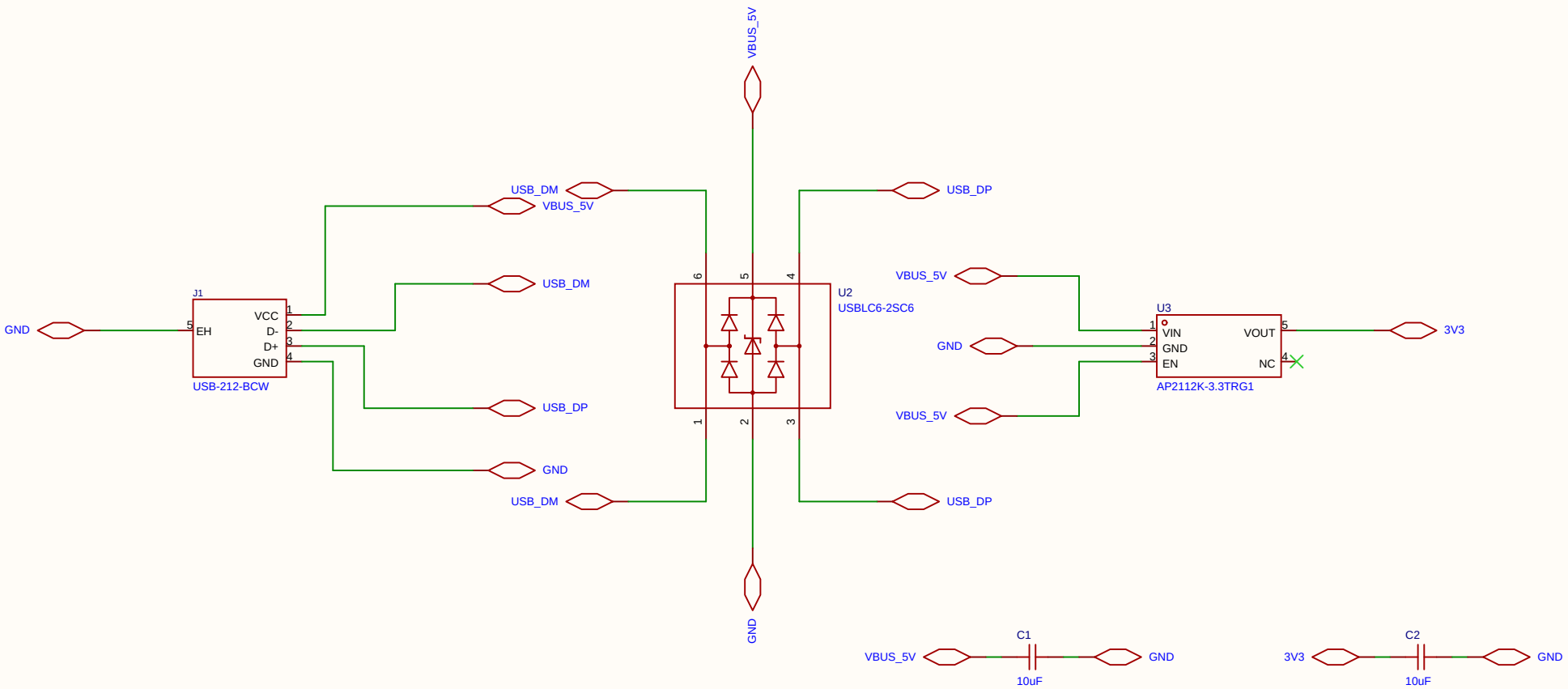
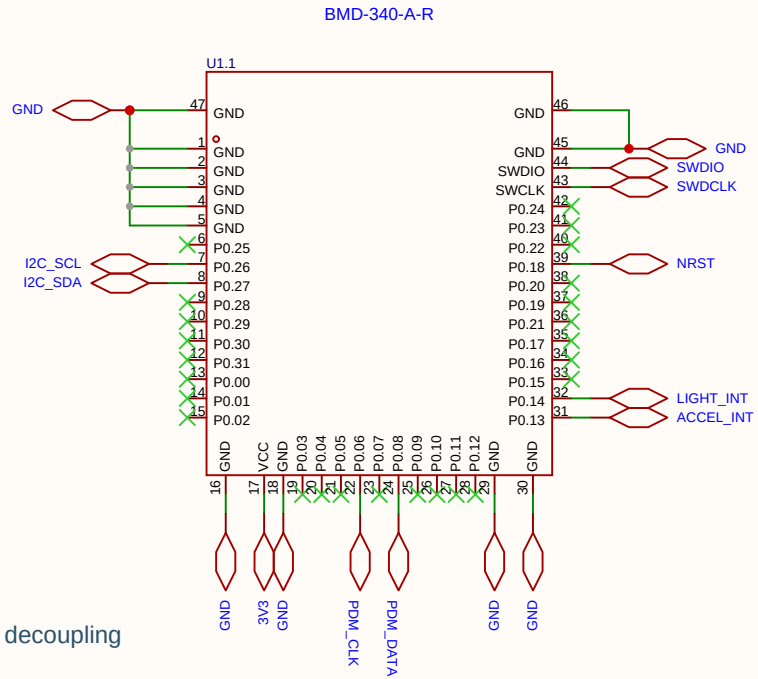


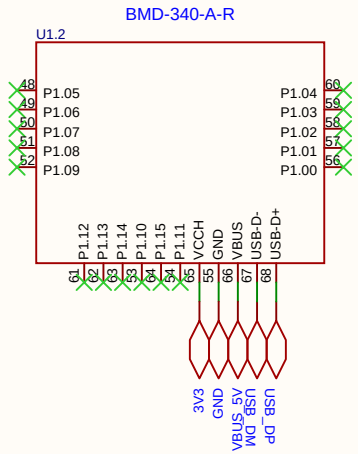
USB Full Speed data connects to the BMD-340 USB peripheral.
USB provides 5 V; AP2112K supplies the 3.3 V domain.

Schematic	Schematic1	Create at	2026-10-10
		Update at	2026-10-17
Board	Board1	Page	01-USB-POWER
Drawn	keisuke054	MODE-USB-ENV-01	
Reviewed			
		Version	Size	Page 1 Total 3
EasyEDA		R1D	A4	keisuke054

BMD-340: application I/O



BMD-340: USB and supply



Module decoupling



Programming pads - PCB underside

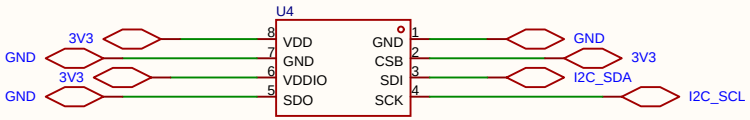
TP1: GND TP2: 3V3 TP3: SWDIO
TP4: SWDCLK TP5: NRST
Signal list only; no physical connector is fitted.

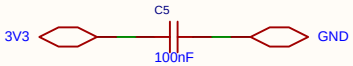
Pin numbers are BMD-340 module pad numbers.
Unused GPIOs are intentionally left open.
Enable the internal pull-up for the open-drain LIGHT_INT signal.

Schematic	Schematic1	Create at	2026-10-10
		Update at	2026-10-17
Board	Board1	Page	02-RADIO
Drawn	keisuke054	MODE-USB-ENV-01	
Reviewed			
		Version / Size	Page 2 Total 3
EasyEDA		R1D / A4	keisuke054

Temperature / humidity / pressure / gas

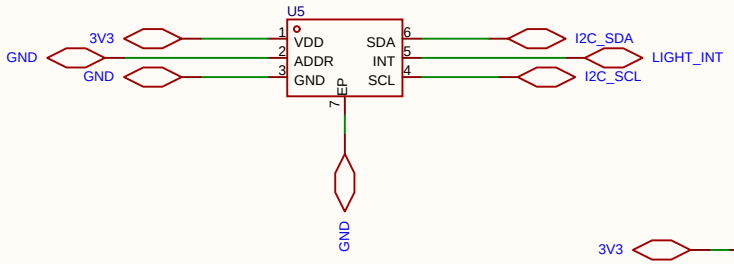
BME688

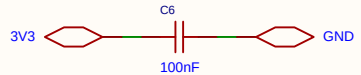




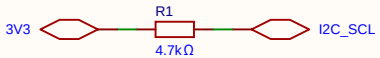
Ambient light

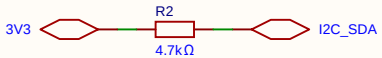
OPT3001DNPR





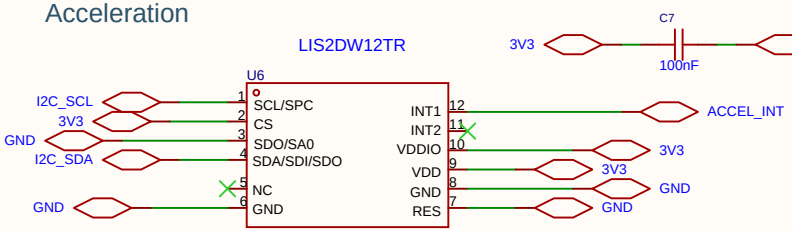
I2C pull-ups

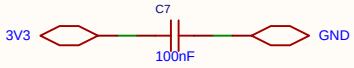




Acceleration

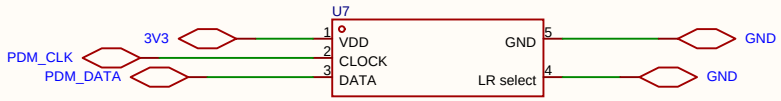
LIS2DW12TR

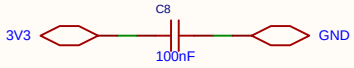




PDM microphone

IM69D128SV01XTMA1





BME688 measures gas resistance, not direct CO2.
Application firmware and physical validation are pending.

Schematic	Schematic1	Create at	2026-10-10
		Update at	2026-10-17
Board	Board1	Page	03-SENSORS
Drawn	keisuke054	MODE-USB-ENV-01	
Reviewed			
		Version	Size	Page 3 Total 3
EasyEDA		R1D	A4	keisuke054